

Optimization of CNN Hyperparameters for Binary Classification

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Mathew Dawson

Introduction and Objectives:

For this project my objective was to create and train a model for binary classification on the PneumoniaMNIST dataset using ResNet18 as a starting point. My approach includes a base training loop which is then run multiple times to find the hyperparameters that result in the model with the best accuracy. I then had to compute some metrics on this model and perform 5-fold cross validation.

Literature Review:

[Optimizing double-layered convolutional neural networks for efficient lung cancer classification through hyperparameter optimization and advanced image pre-processing techniques](#) - This paper provided valuable insight on metrics and how they relate to the performance of the model, as well as loss functions.

[The Role of Hyperparameter Optimization in Fine-Tuning of Cnn Models](#) - This paper provided valuable insight on the hyperparameters and which are most important for optimisation.

Technical Details:

In order to use the PneumoniaMNIST dataset with ResNet18, I had to modify the input and output layers, as ResNet18 by default takes 3 channel RGB images while the dataset has single channel grayscale images.

During hyperparameter tuning, the following hyperparameters are tuned -

- Learning rate - [1e-1 to 1e-5]
- Optimizer - [Adam, SGD]
- Weight decay - [1e-3 to 1e-10]

As I used the Optuna hyperparameter tuning library, learning rate and weight decay are evaluated along a logarithmic curve rather than a linear one. Optuna also provides functionality for “pruning” where models that perform poorly are stopped early, which saves some time when tuning.

My implementation features set seeds for reproducibility

Evaluation:

Running my hyperparameter training loop for 100 iterations results in a model with 0.85 accuracy and 0.96 auc, with 0.93 average accuracy when doing 5 fold cross validation. This is above baseline.

I also achieved:

- AUPR: 0.96
- Precision: 0.82
- Recall: 0.99
- F1 Score: 0.89

My model performs better than baseline due to being finely tuned for better performance, by elimination of poorly performing models. The model is shown to perform better when doing 5 fold cross validation, which is arguably a more accurate method of evaluation.

